

# Secure Authentication System

**Internship:** Growfinix

**Student:** Divyadharshini M

## Abstract

This project implements a secure user authentication system using React, Node.js, Express, MongoDB Atlas, JWT, bcrypt, and Tailwind CSS. Users can register, log in securely, access a protected dashboard, and log out safely.

## Problem Statement

Many web applications require secure authentication to protect user accounts and sensitive data. This project demonstrates a secure login and registration system using modern web technologies.

## Objectives

- Develop a secure authentication system.
- Protect passwords using bcrypt.
- Implement JWT-based authentication.
- Connect the application with MongoDB Atlas.
- Build a responsive user interface using React and Tailwind CSS.

## Technologies Used

Frontend: React, Vite, Tailwind CSS

Backend: Node.js, Express.js

Database: MongoDB Atlas

Authentication: JWT, bcrypt

Tools: Git, GitHub, Postman

## Modules

1. User Registration
2. User Login
3. JWT Authentication
4. Protected Dashboard
5. Logout

## Project Structure

Task-1-Secure-Authentication-System/

```
|— client
|— server
|— screenshots
|— README.md
```

## Installation

1. Install dependencies.
2. Configure .env.
3. Start backend using `npm run dev`.
4. Start frontend using `npm run dev`.
5. Open `http://localhost:5173`.

## Future Enhancements

Email verification, password reset, role-based access control, OAuth login, profile management.

## Conclusion

The project successfully demonstrates a complete full-stack authentication system following secure development practices.